The Legacy and Continued Relevance of the *Hubble Space Telescope*

Dr. Rolf A. Jansen (ASU/SESE, Research Scientist)

Image credit: NASA, ESA, N. Smith

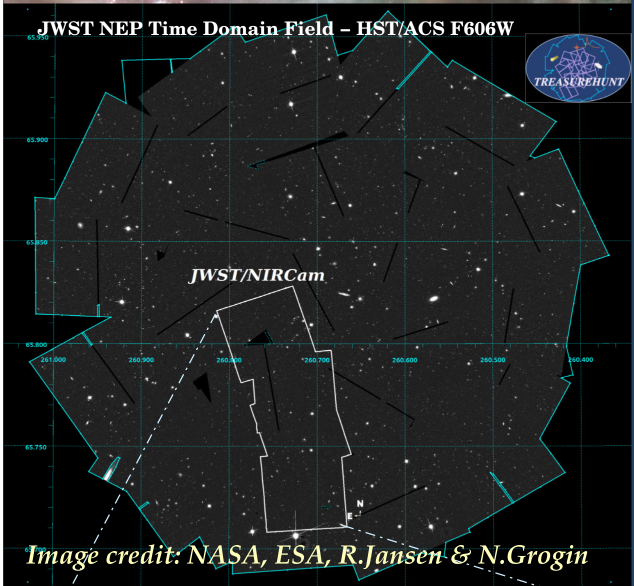
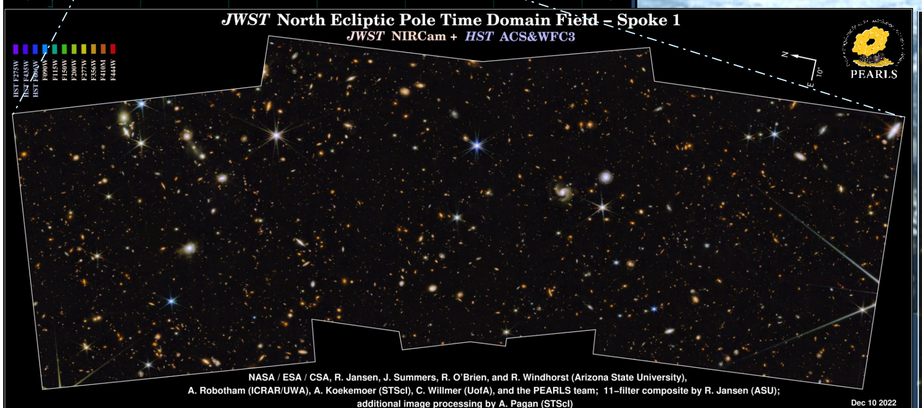
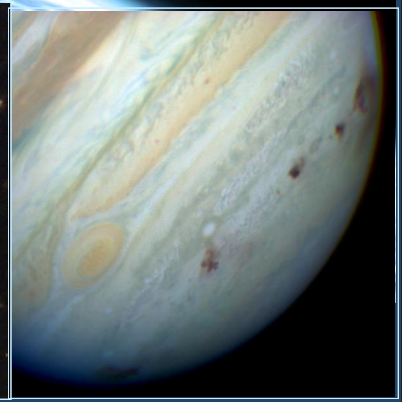


Image credit: NASA, ESA, R.Jansen & N.Grogin

Image credit: NASA





The Legacy and Continued Relevance of the Hubble Space Telescope

- **Once Upon A Time..**

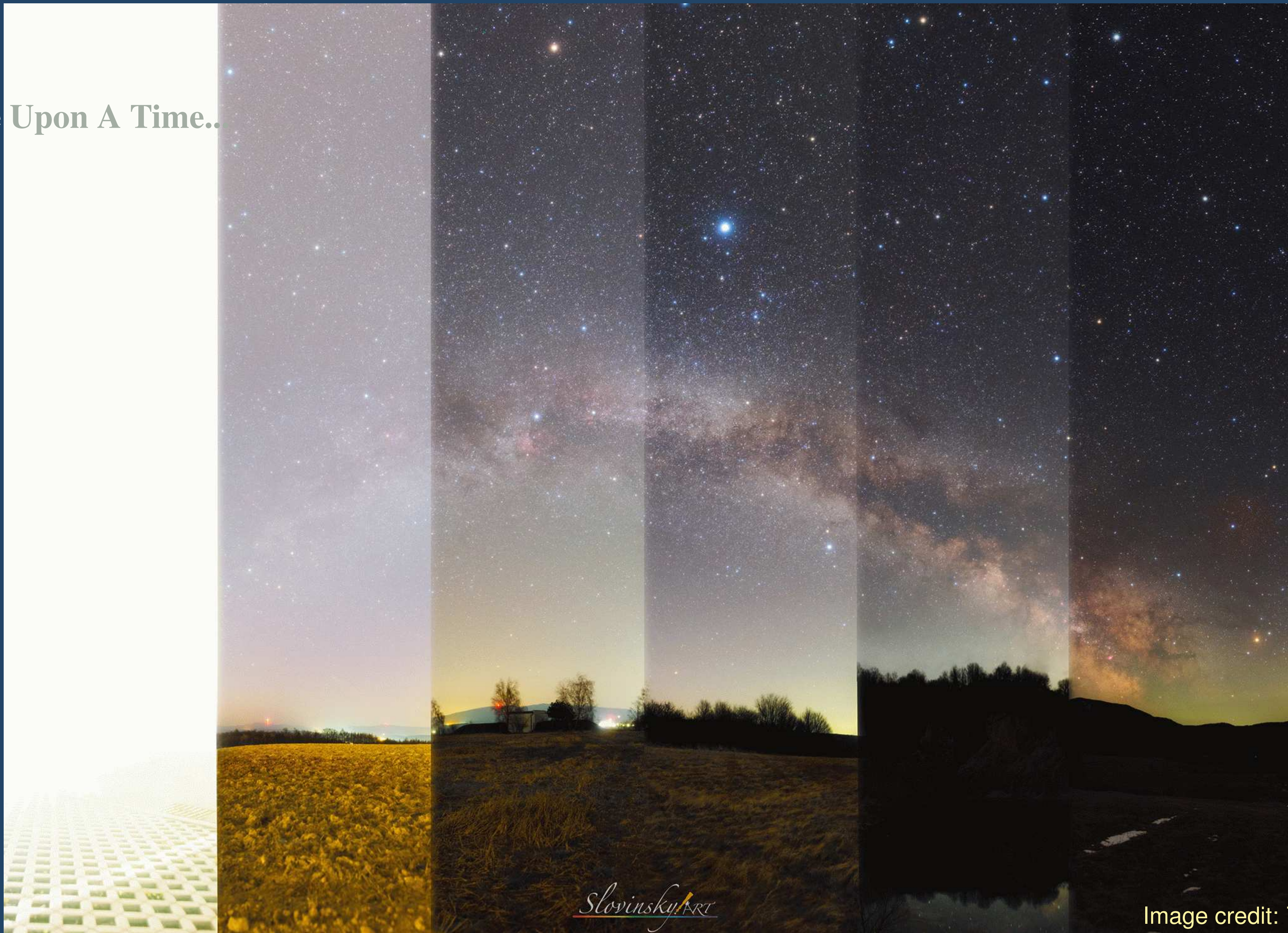


Image credit: T. Slovinsky

The Legacy and Continued Relevance of the Hubble Space Telescope

- **Once Upon A Time...**

The Legacy and Continued Relevance of the Hubble Space Telescope

- **Before the *Hubble* revolution...**
  - **stark photographic images**

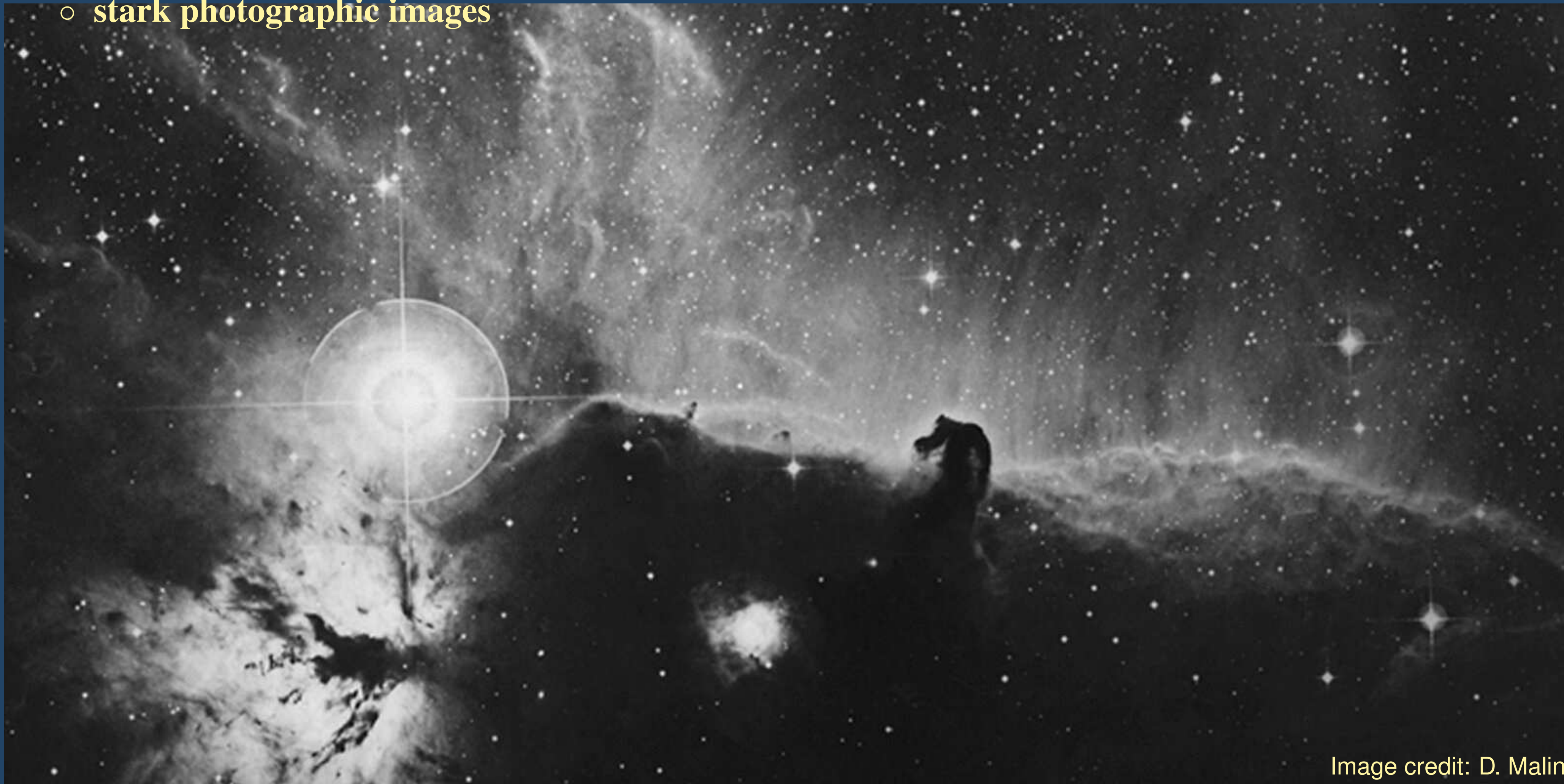


Image credit: D. Malin

The Legacy and Continued Relevance of the Hubble Space Telescope

- **Before the *Hubble* revolution...**
  - **stark photographic images**
  - **Earth's atmosphere blocks light outside the visible range and blurs images**

The Legacy and Continued Relevance of the Hubble Space Telescope

- After the *Hubble* revolution...
  - *Hubble* provides darker sky background
  - higher resolution
  - access to wavelengths of light inaccessible from the ground

Credit: NASA, ESA, Hubble Legacy Archive
Processing & Copyright: R. Bernal Andreo

Hubble's Galactic Legacy

- *Hubble*'s "Pillars of Creation" – stellar birth
  - massive clouds of gas and dust are being eroded away by young, massieve stars
  - new low-mass stars are still being formed, breaking out of pillar tips

Credit: NASA, ESA, *Hubble*; J. Hester amp; P. Scowen (ASU)

Credit: NASA, ESA, *Hubble*, HLA; processing: L. Romero

Hubble's Galactic Legacy

- **The Great Orion Nebula – Trapezium region**
  - **nearest stellar nursery to our Sun, several very massive O-stars in the Trapezium Cluster**

Credit: NASA, ESA, M. Robberto (STScI/ESA) and the HST Orion Treasury Team

Hubble's Galactic Legacy

- **The Carina Nebula – Orion raised to another level**
  - ○ **multiple episodes of massive star formation and many pilars of creation**
  - ○ **many of the most massive stars in our Galaxy**

Credit: NASA, ESA, N. Smith (UC Berkeley), Hubble Heritage Team (STScI/AURA)

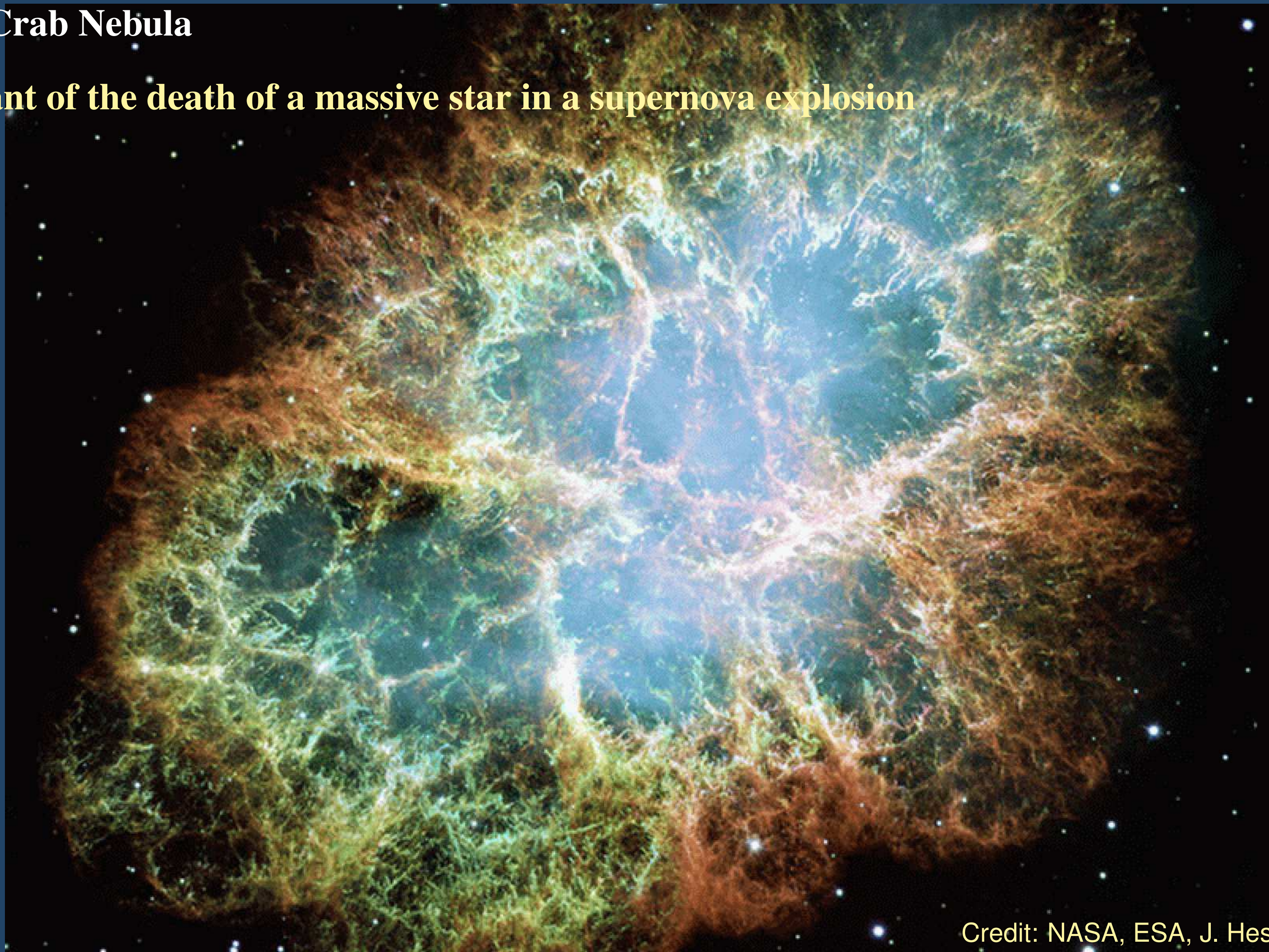
Hubble's Galactic Legacy

- **Eta Carina – precursor to stellar death**
  - **late stage of massive stellar evolution**

Credit: NASA, ESA, N. Smith (UC Berkeley), Hubble Heritage Team (STScI/AURA)

Hubble's Galactic Legacy

- **M1: The Crab Nebula**
  - **remnant of the death of a massive star in a supernova explosion**



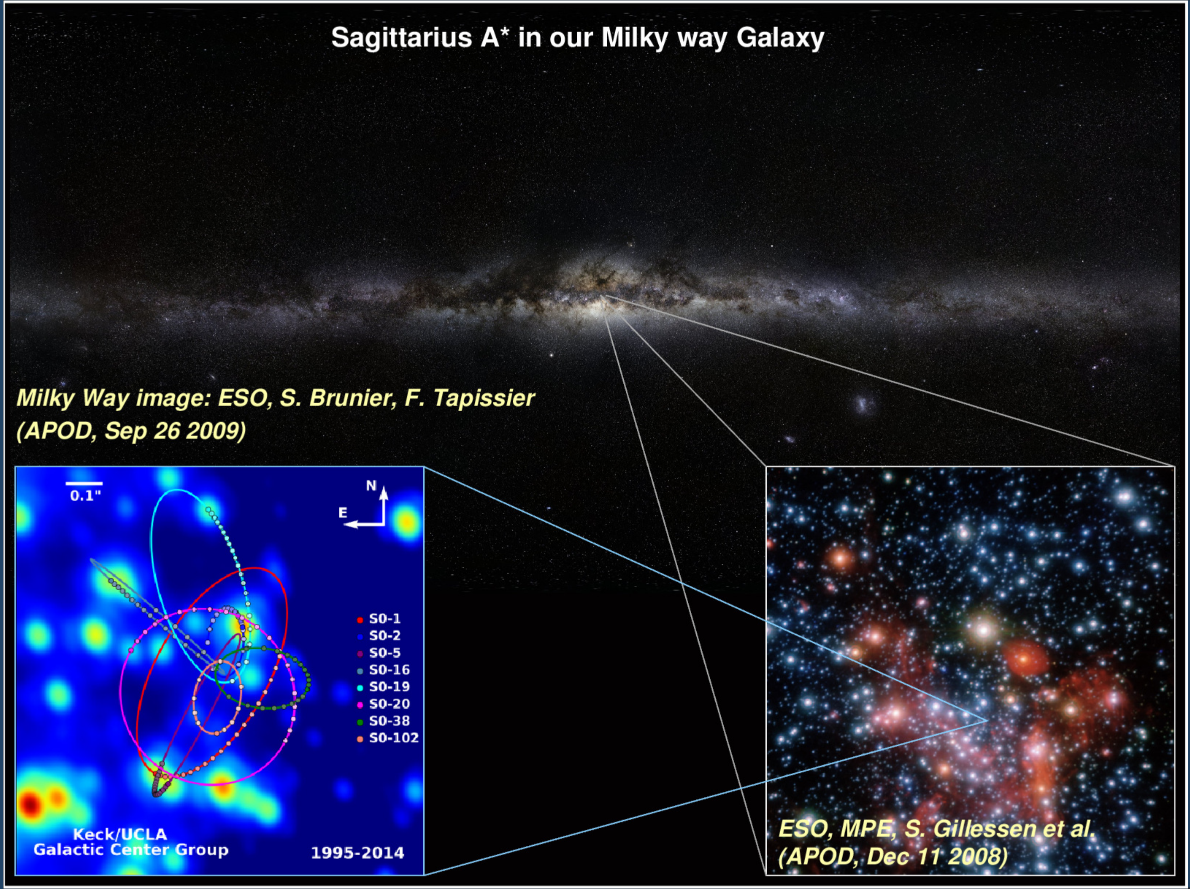
Credit: NASA, ESA, J. Hester & A. Loll (ASU)

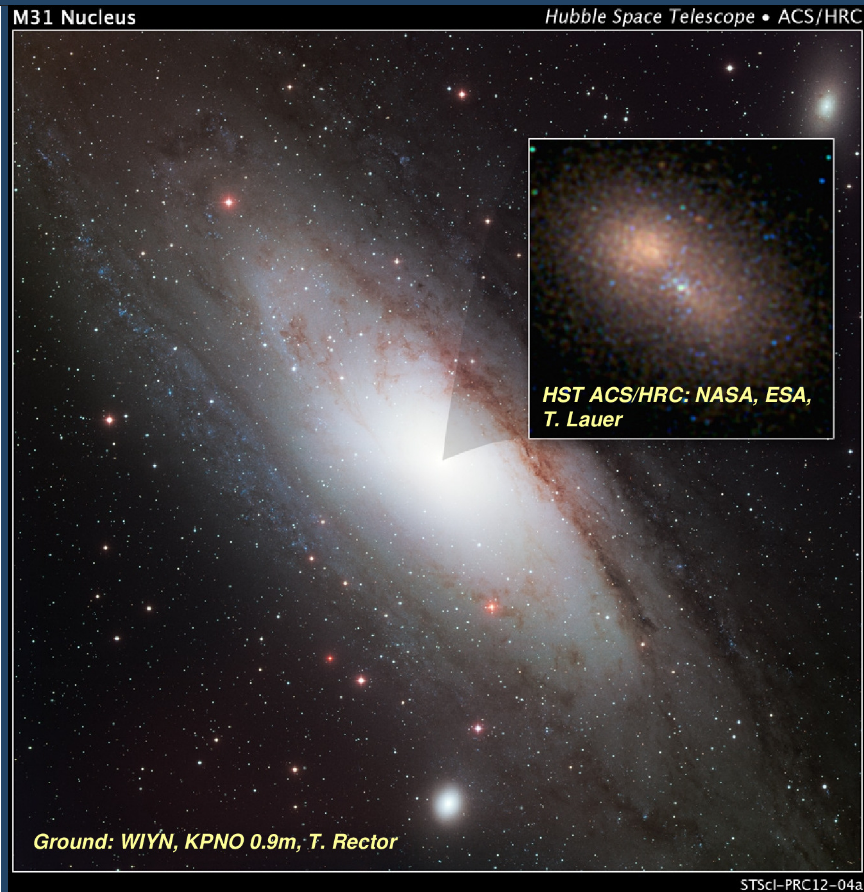
Hubble's Extragalactic Legacy

- Moving beyond our own Milky Way Galaxy...
  - *Hubble* discovered that all large galaxies have a super-massive black hole at their center...
  - growth of that SMBH somehow kept pace with the build-up of stellar mass over cosmic time

Hubble's Extragalactic Legacy

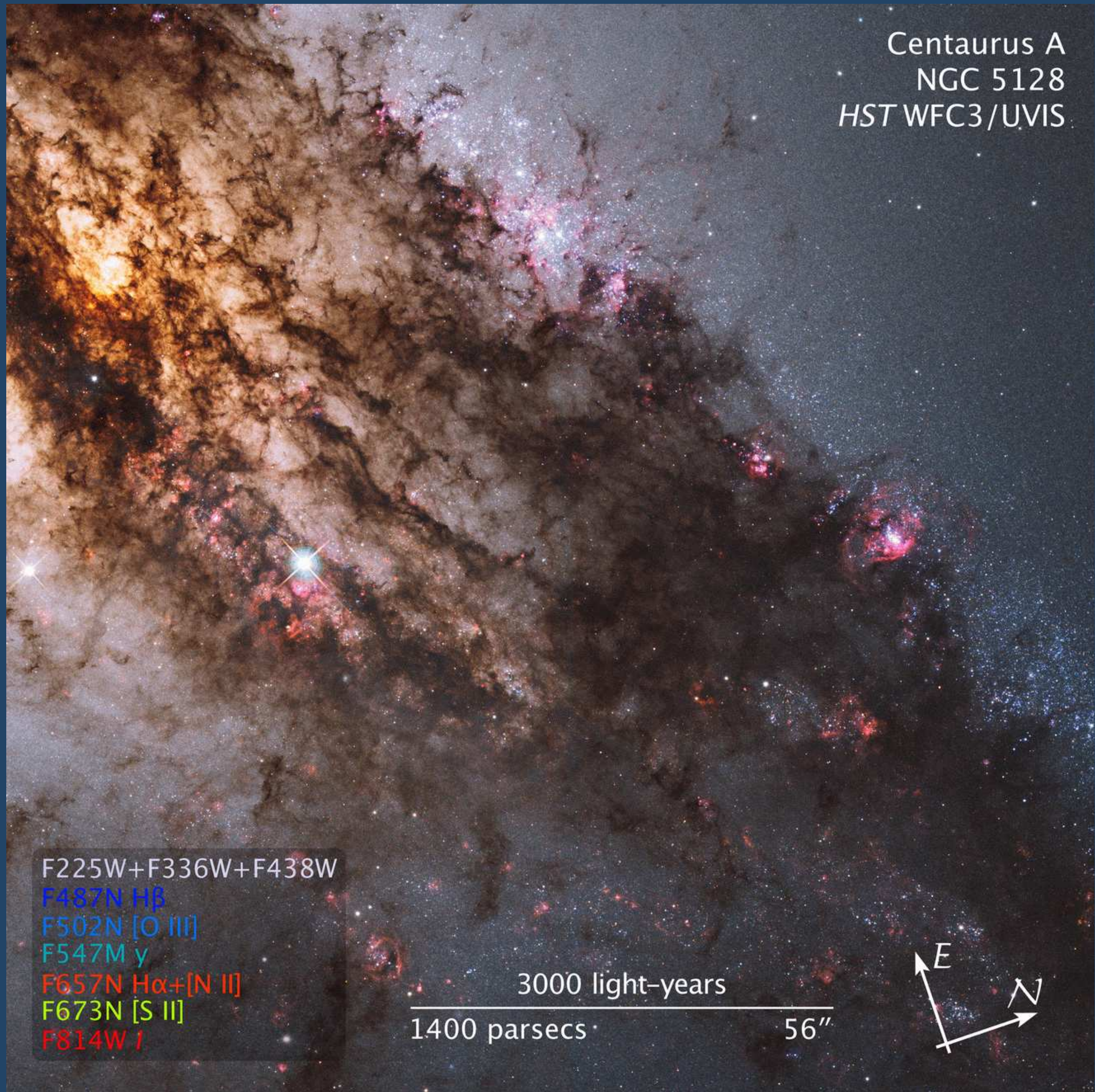
- After the *Hubble* revolution...
  - ... even our own Milky Way Galaxy and our nearest big neighbor, M 31, the Andromeda Galaxy have a supermassive black hole at their center





Hubble's Extragalactic Legacy

- *Hubble* gave us sharp views of starbursting galaxies...



Credit: NASA, ESA, Hubble Heritage Team (STScI/AURA)–ESA/Hubble Collaboration; R. O'Connell et al.

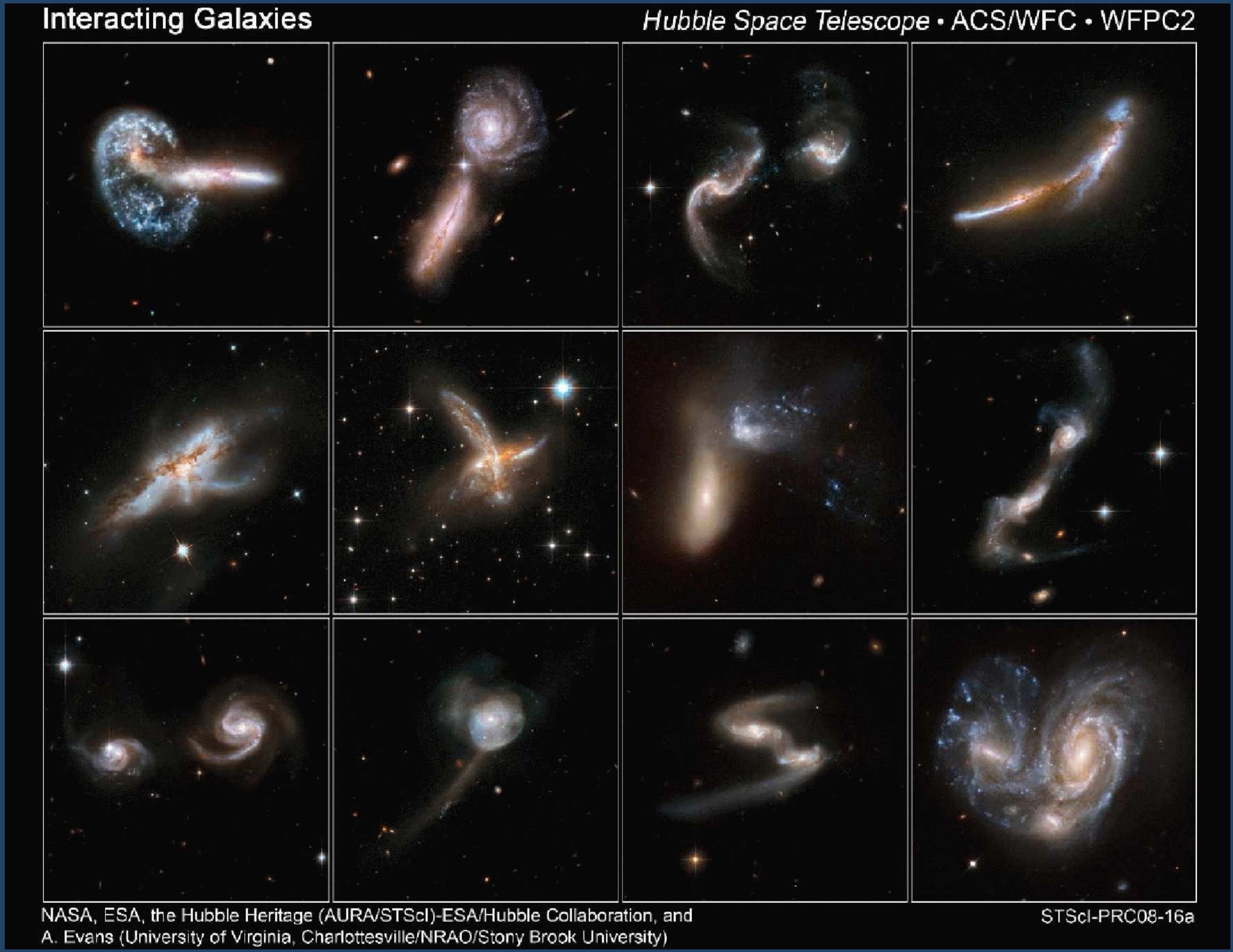
Hubble's Extragalactic Legacy

- *Hubble* gave us sharp views of interacting and merging galaxies...

Credit: NASA, ESA, Hubble Heritage Team (STScI/AURA)–ESA/Hubble Collaboration; B. Whitmore & J. Long

Hubble's Extragalactic Legacy

- ... and allowed us to trace galaxy assembly and subsequent evolution over the past 12 billion years



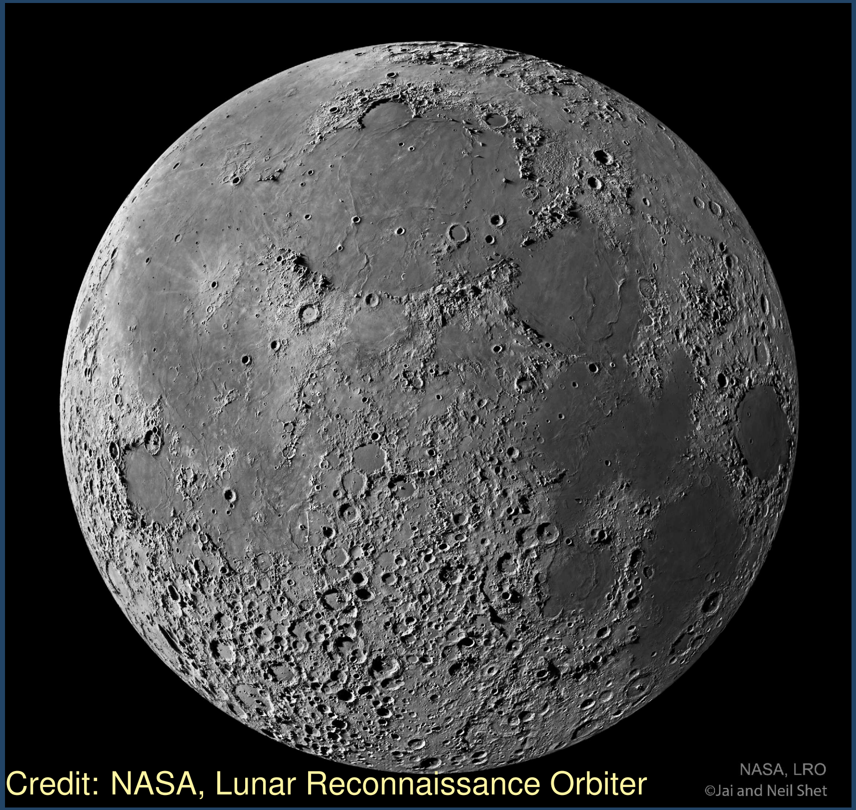
Hubble's Extragalactic Legacy

- **12 billion years of cosmic history in one *Hubble* picture**
  - **galaxies at redshifts larger than 7 are hard to find (dim, red, and tiny), and very rare**

Credit: NASA, ESA, R. Windhorst (ASU), P. McCarthy (CIW), & R. O'Connell (UVa); color composite by M. Mechtley (ASU)

Hubble's Solar System Legacy

- Before *Hubble* ...
  - evidence of a violent Solar System past
  - uncertain present-day rate of impacts, and types of impactor

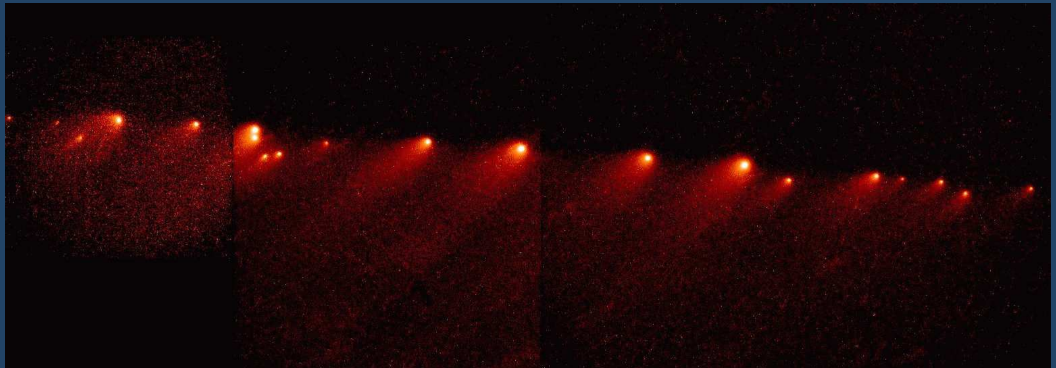


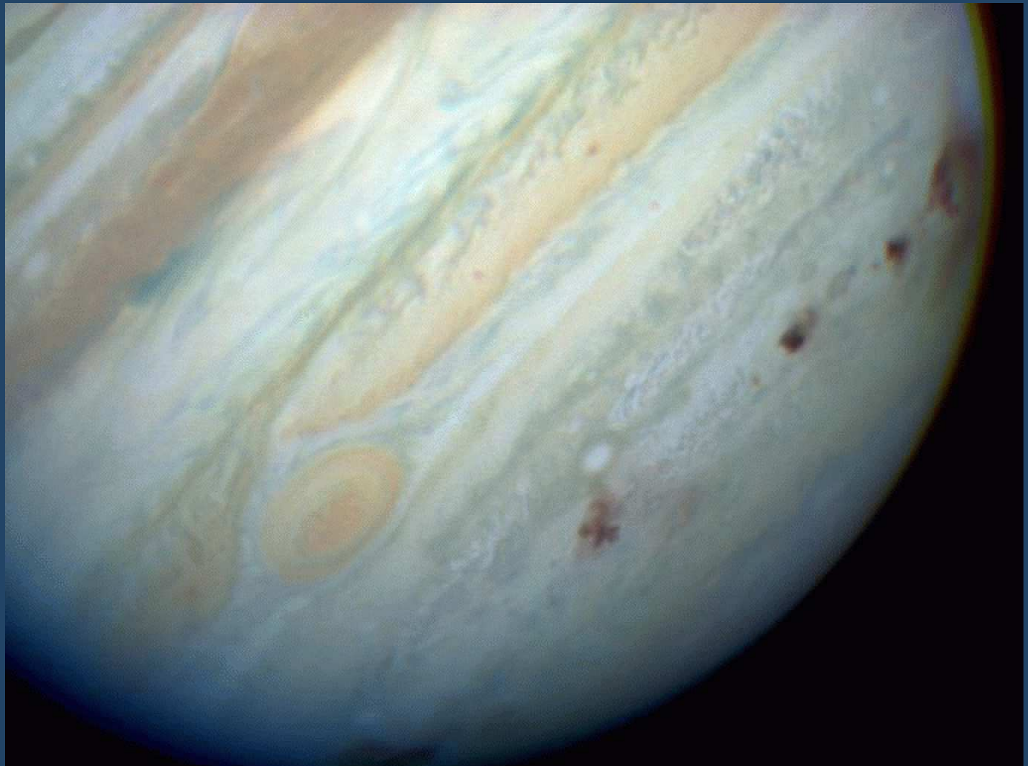
Credit: NASA, Lunar Reconnaissance Orbiter


Credit: USGS

Hubble's Solar System Legacy

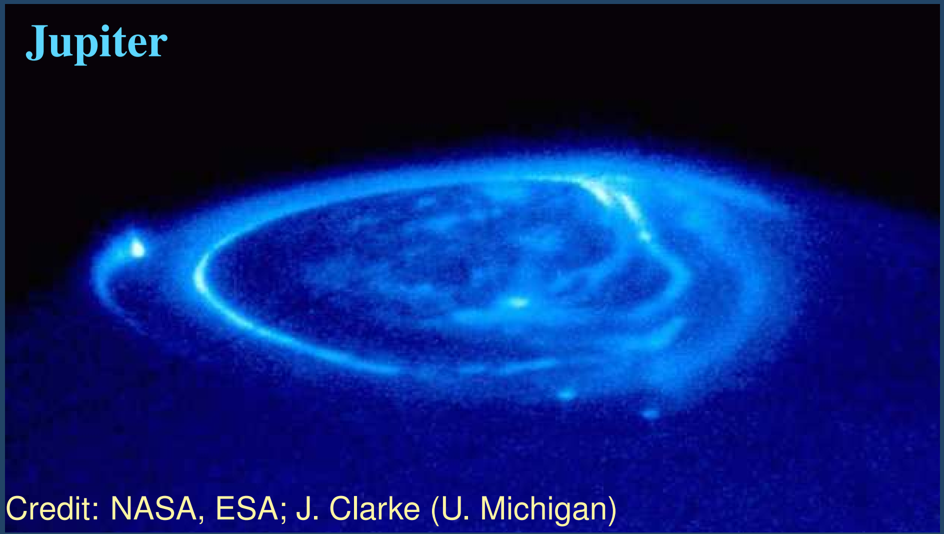
- *Hubble* observed 1992 break-up of a comet, and subsequent impact of its pieces on Jupiter in July 1994
  - the process is ongoing
  - Jupiter acts as the Solar System's giant hoover

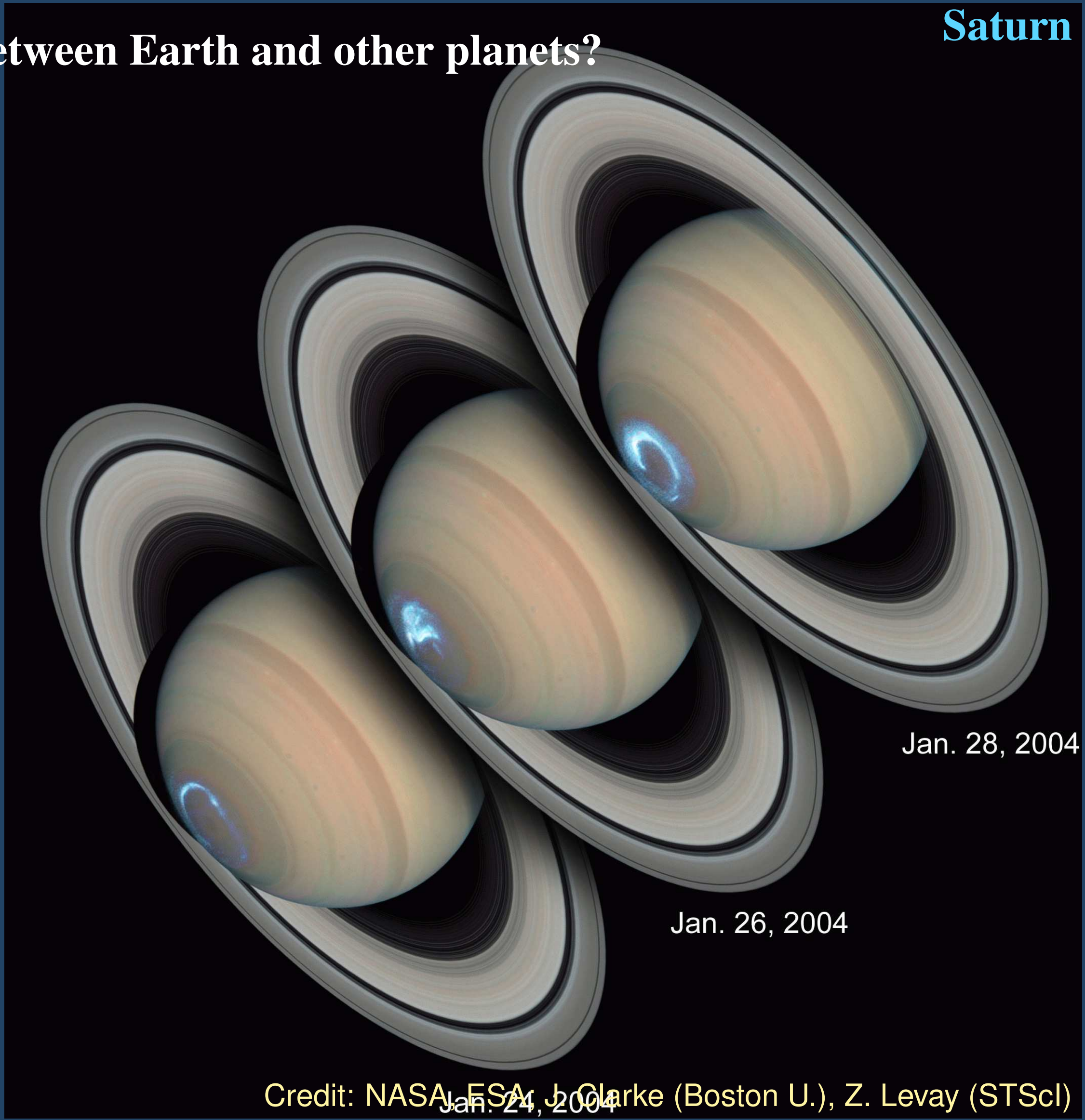

Credit: NASA, ESA; H. Weaver (JHU), T. Smith (STScI)


Credit: NASA, ESA, *Hubble*

Hubble's Solar System Legacy

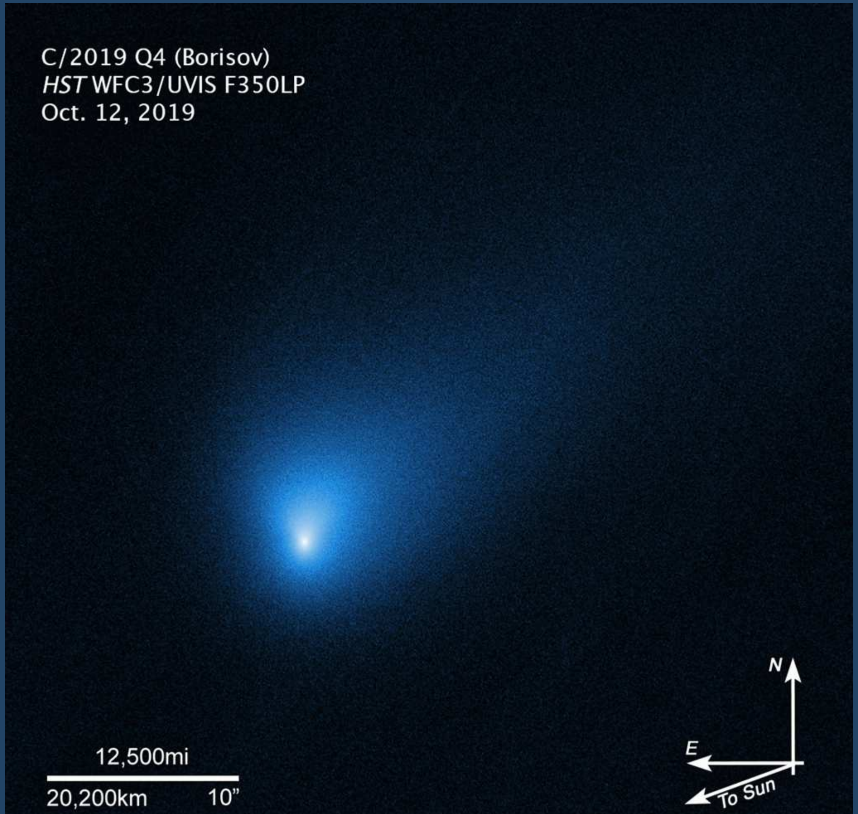
- **Do magnetic fields and space weather differ between Earth and other planets?**
  - ***Hubble* observed persistent aurorae around Saturn's and Jupiter's poles**
  - **allowed comparing similarities and differences of Earth's and Saturn's magnetic fields, and interactions with the solar wind**



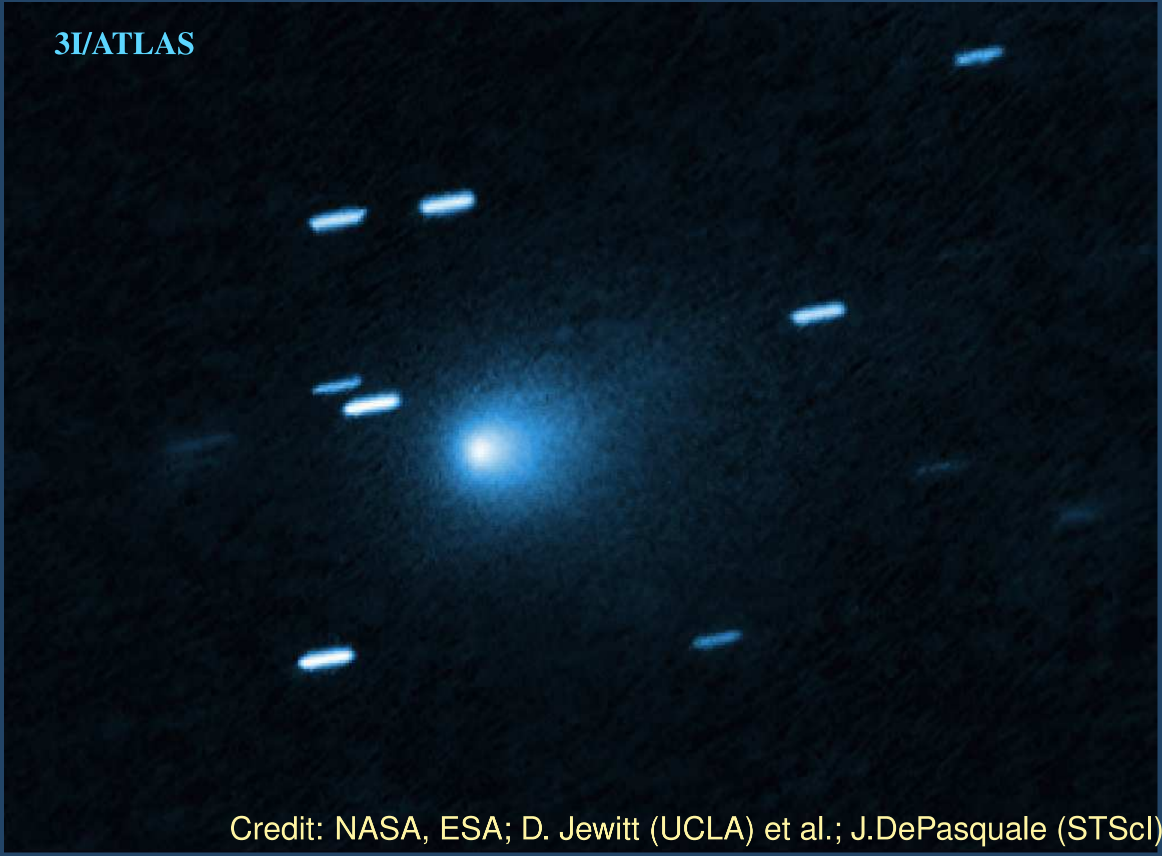


Hubble's Solar System Legacy

- **Visitors from Beyond Our Solar System?**
  - ***Hubble* observed interstellar interloper 2I/Borisov and 3I/ATLAS**
  - **as they warmed on their approach to the Sun, both shed copious amounts of gas and dust**



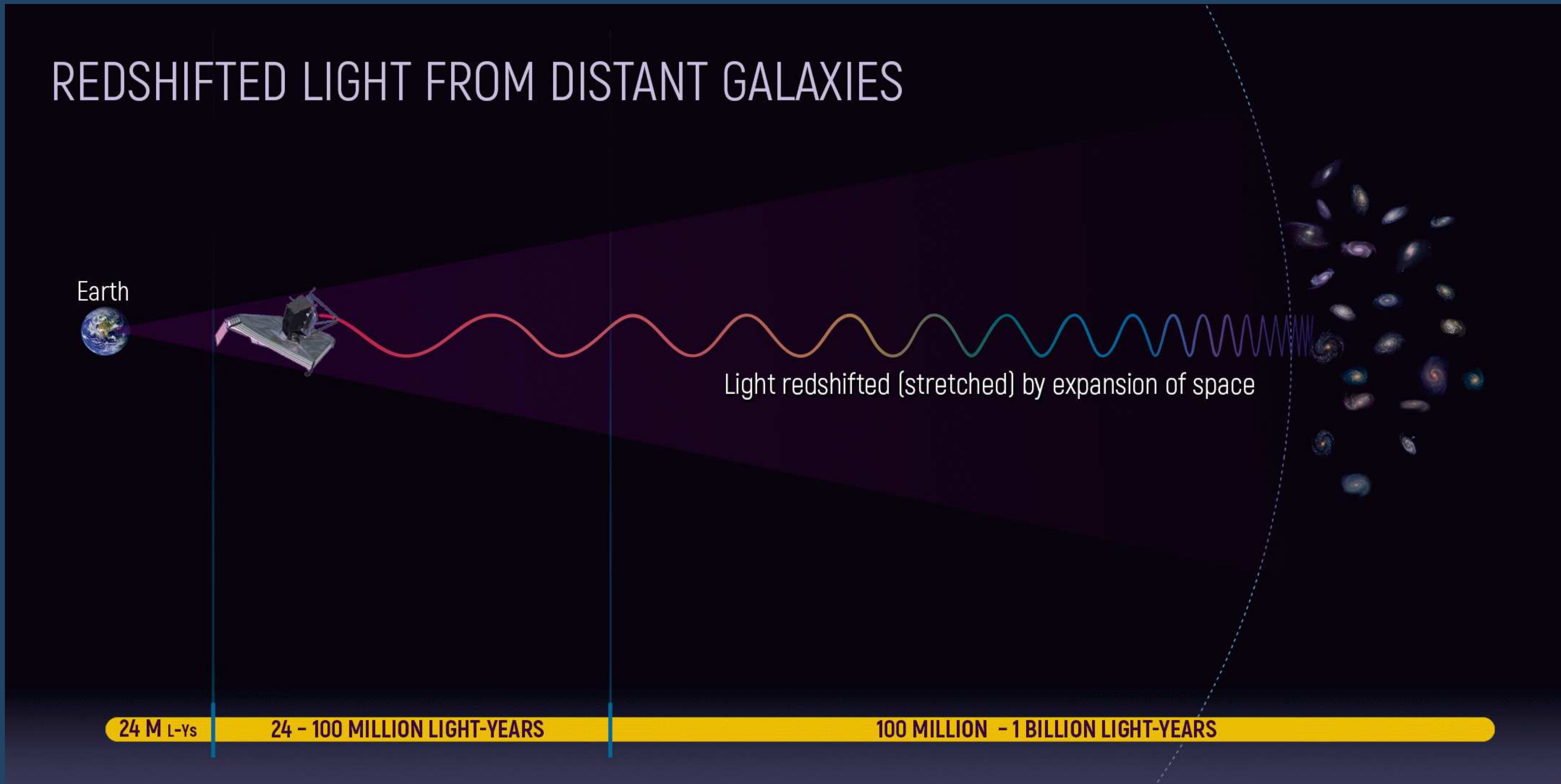
Credit: NASA, ESA; D. Jewitt (UCLA) et al.



Credit: NASA, ESA; D. Jewitt (UCLA) et al.; J.DePasquale (STScI)

Beyond Hubble?

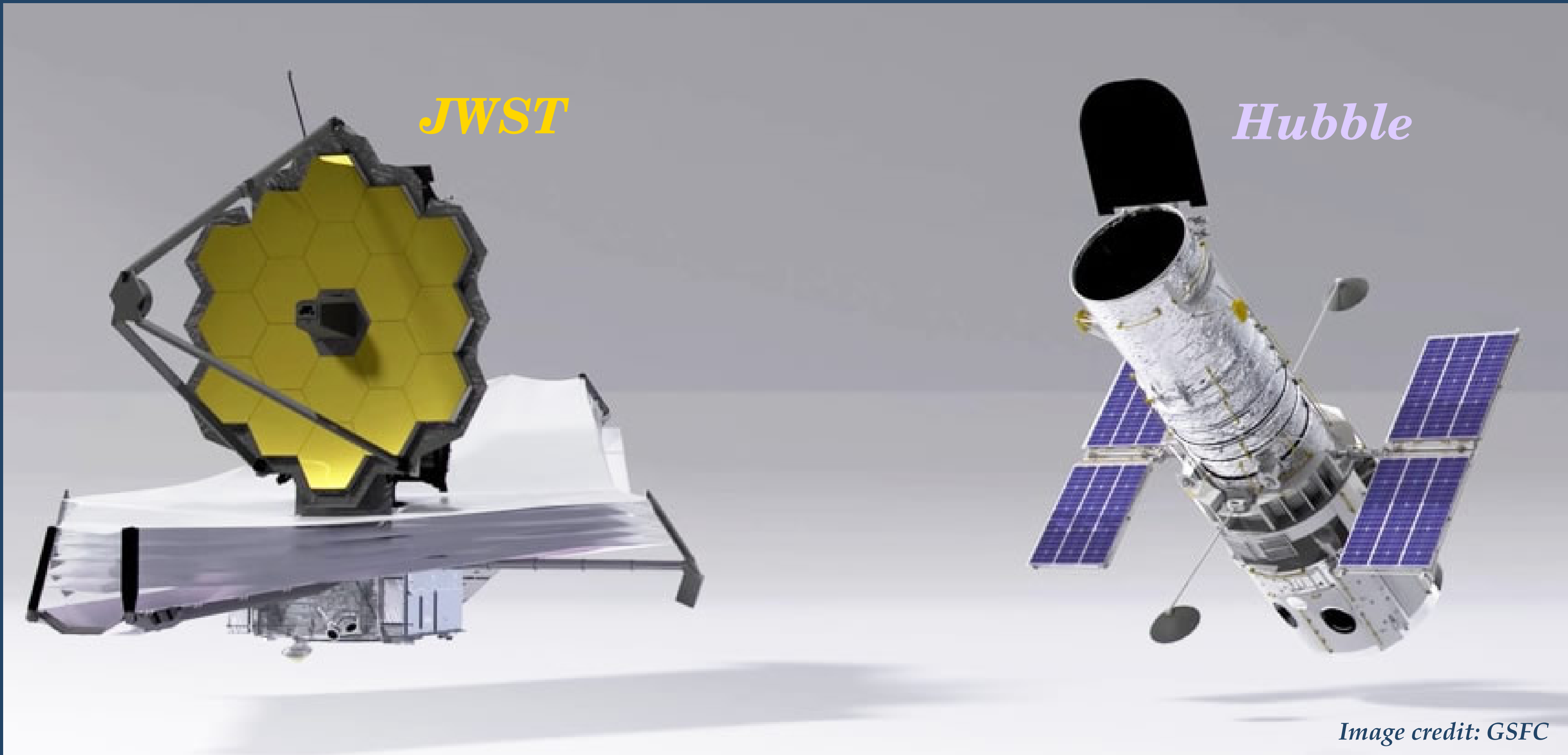
- the wavelengths of light emitted by distant galaxies gets stretched as it travels to us.
  - ultraviolet–visible light from the first galaxies in the Universe arrives as *infrared* light



Credit: NASA and A. Feild (STScI)

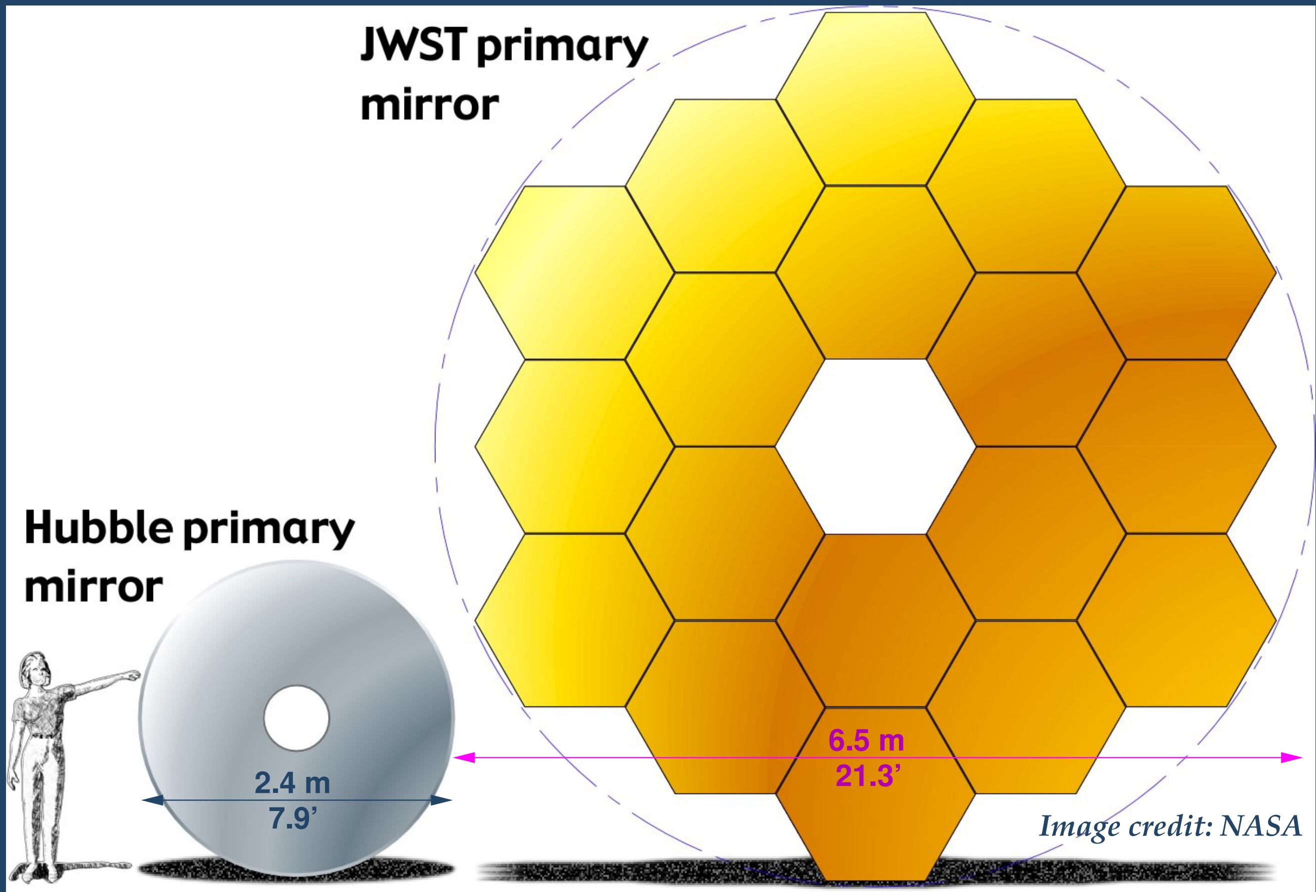
Beyond Hubble

- *JWST* is the scientific successor of *Hubble*



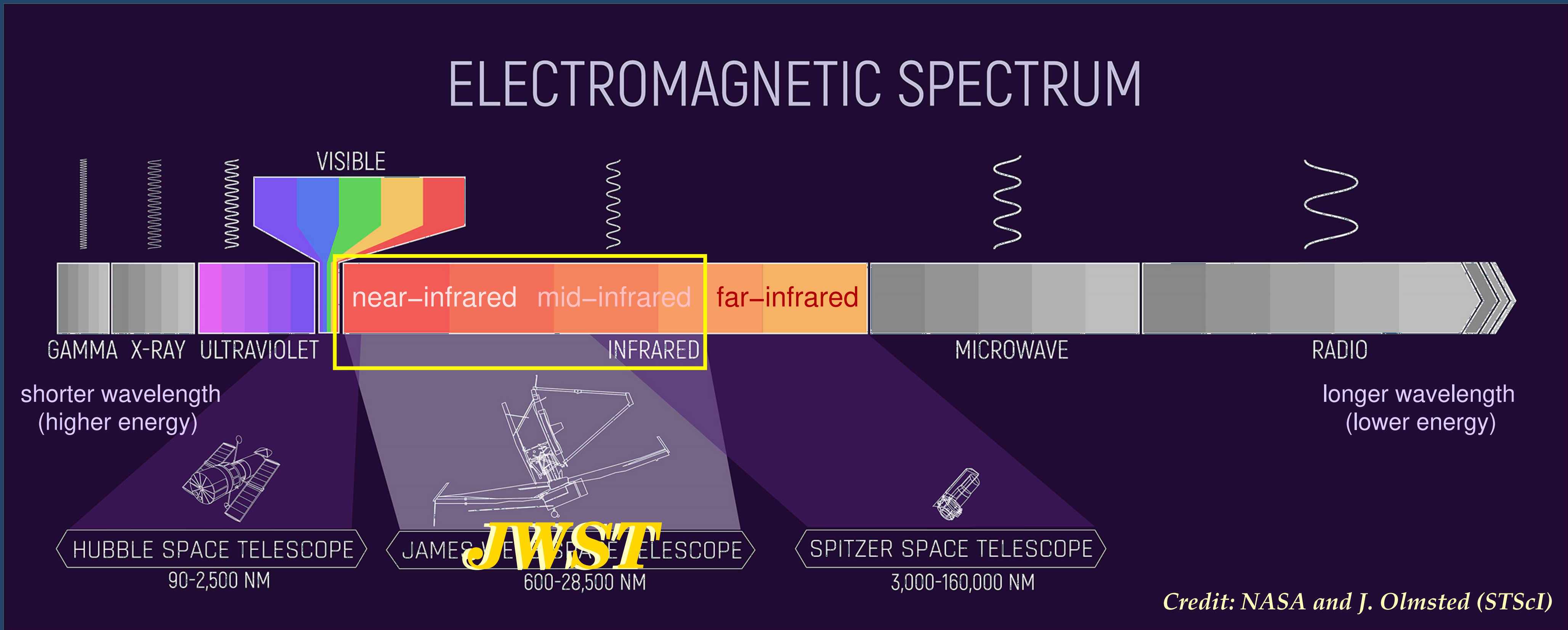
Beyond Hubble

- *JWST*'s primary mirror spans 6.5 m in diameter
  - 2.7 times wider than *Hubble*'s mirror
  - 6.25 times greater light collecting area



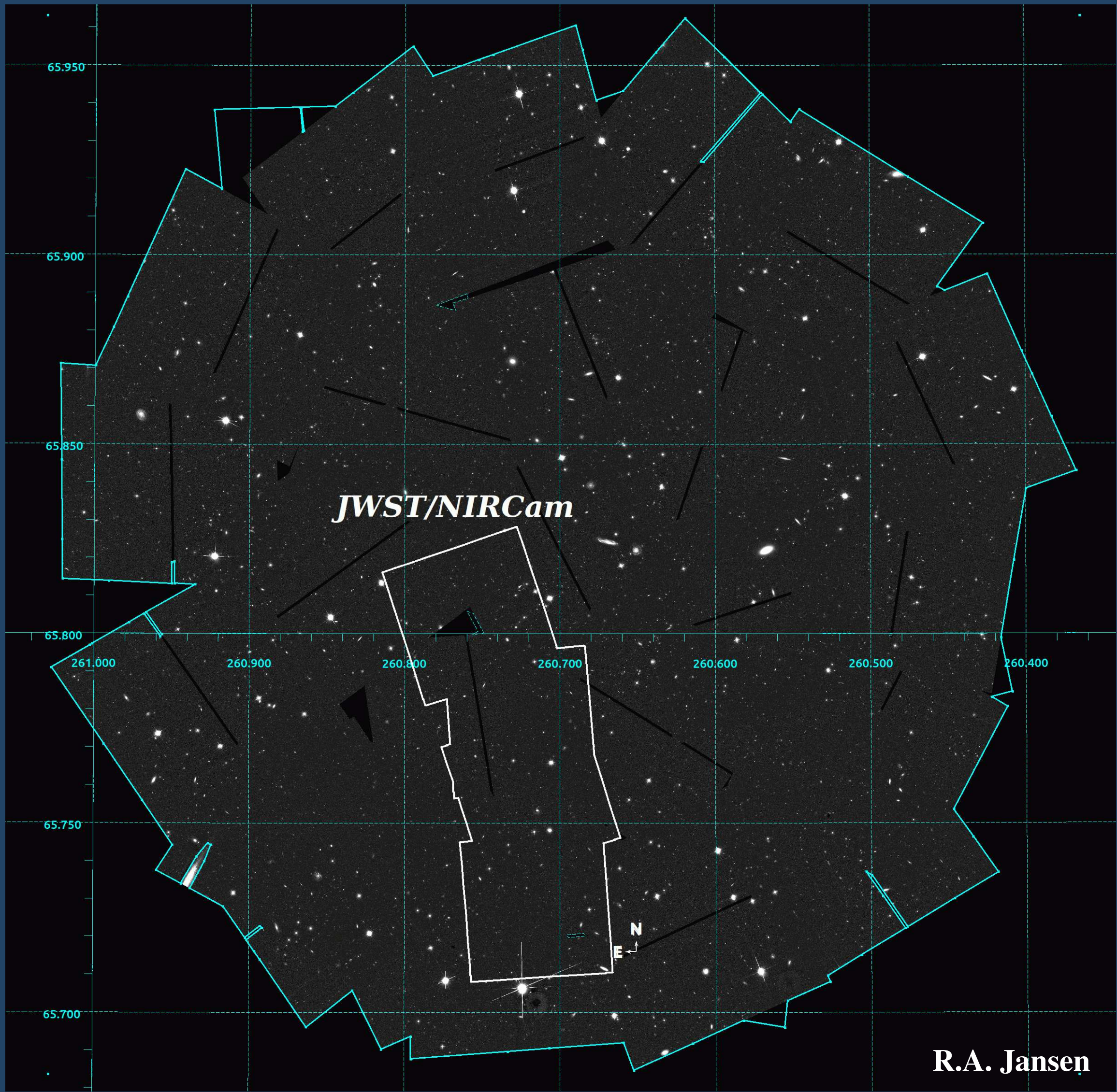
Beyond Hubble

- *JWST* sees farther into the infrared than *Hubble*
- *JWST* has a larger mirror to collect the dim light of the first galaxies
- *JWST* has a larger mirror to create sharper images of such small objects.



Credit: NASA and J. Olmsted (STScI)

First Near-Infrared observations in the NEP TDF with *JWST*



- **Footprint of *JWST*/NIRCam "Spoke 1" in the *JWST* NEP Time-Domain Field as executed Aug 28 and Sep 14 2022 in 8 broad-band filters (F090W through F444W) as part of *JWST* PEARLS program GTO 2738 (PI: R. Windhorst), overlaid on a *HST* ACS/WFC F606W mosaic from the TREASUREHUNT program.**

- **Loss of guide star resulted in a difference in orientation between the first and second pointing in the 2$\times$1 mosaic.**

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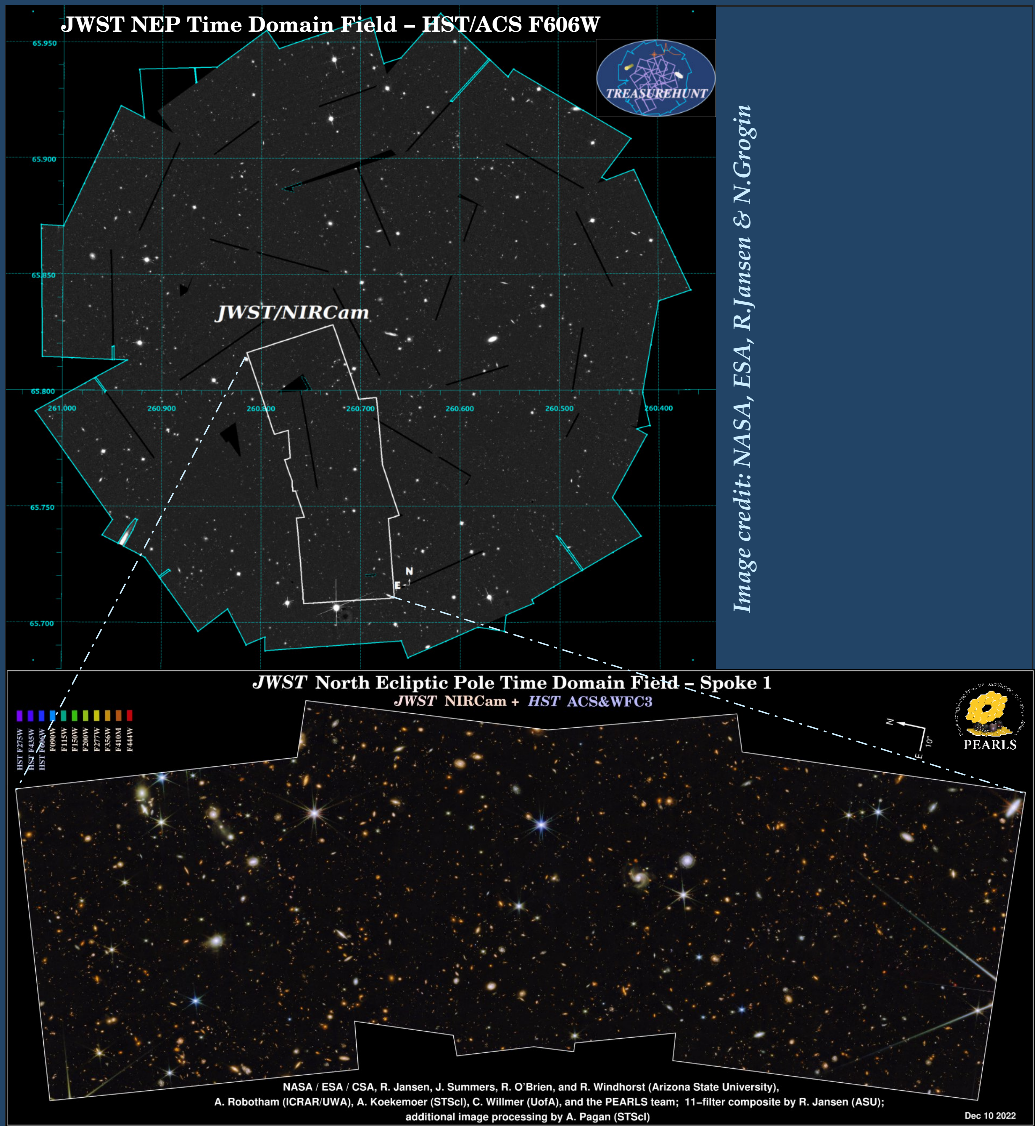

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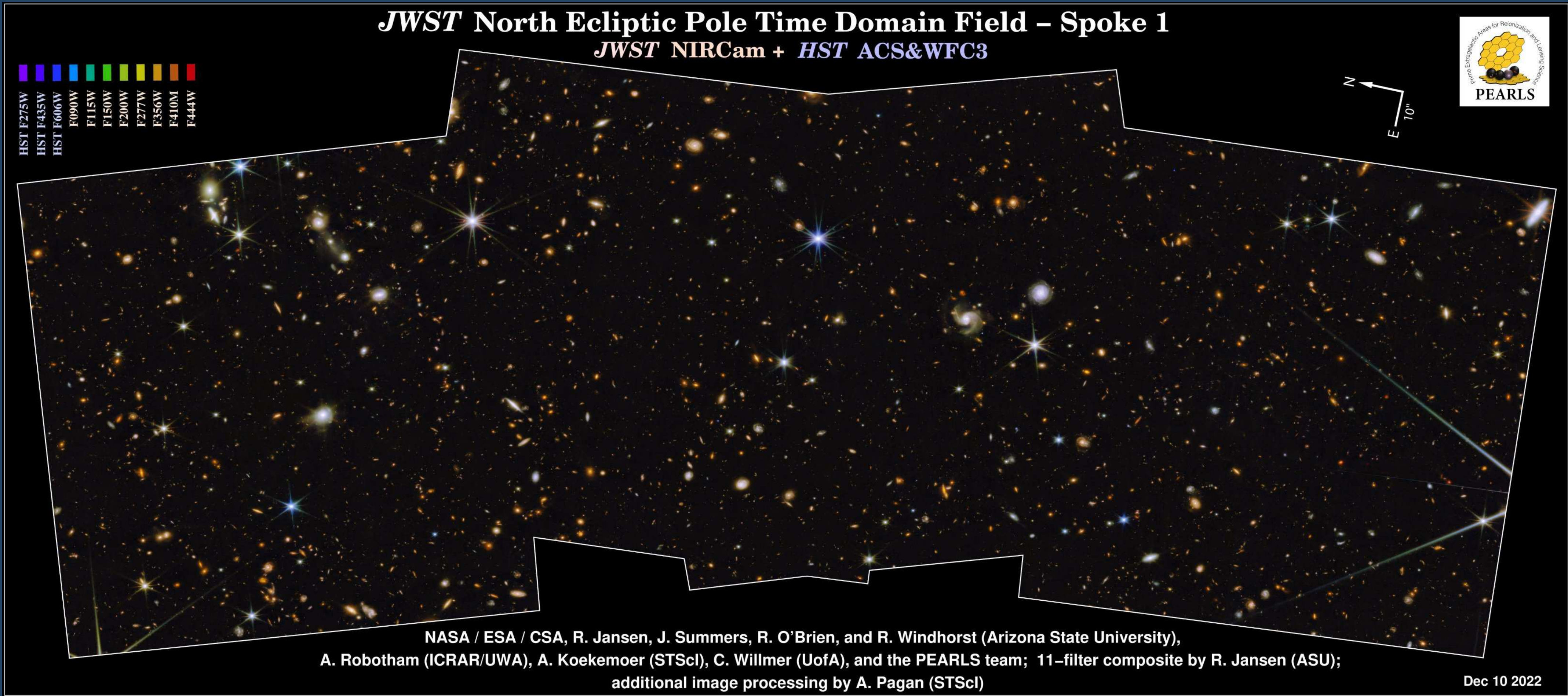
- **Rotating vertical "Spoke 1" to a horizontal orientation...**

First Near-Infrared observations with *JWST*



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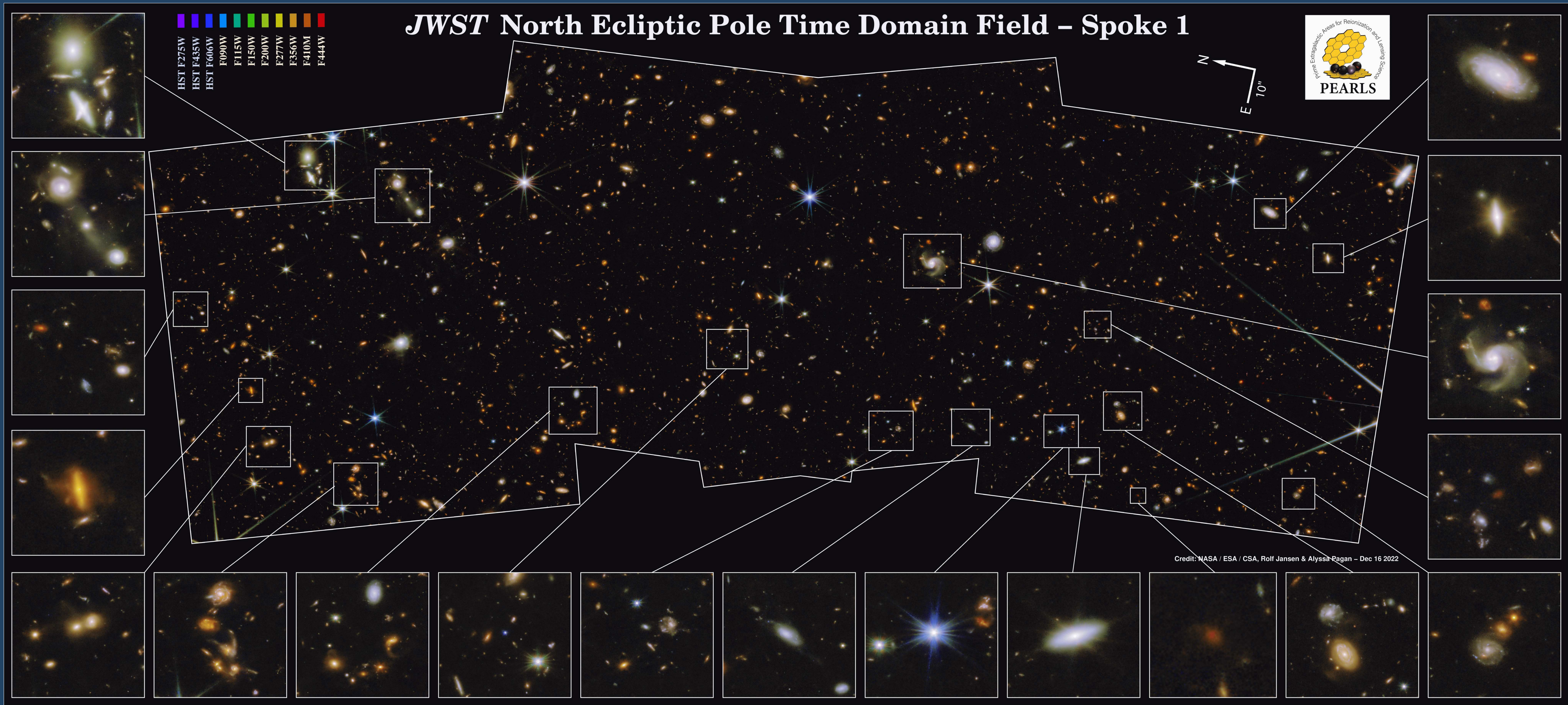

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HST & *JWST* Explorations of a new Time-Domain and Deep Field

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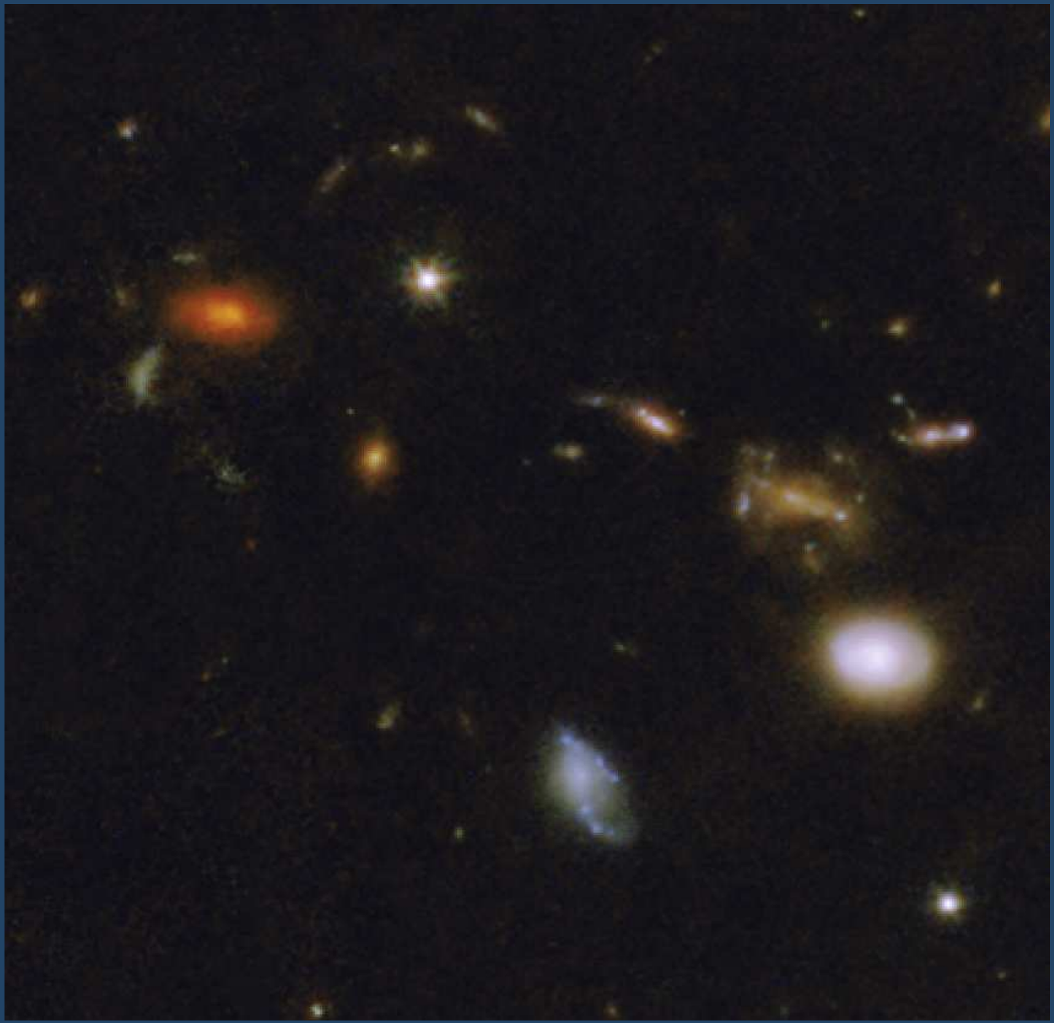

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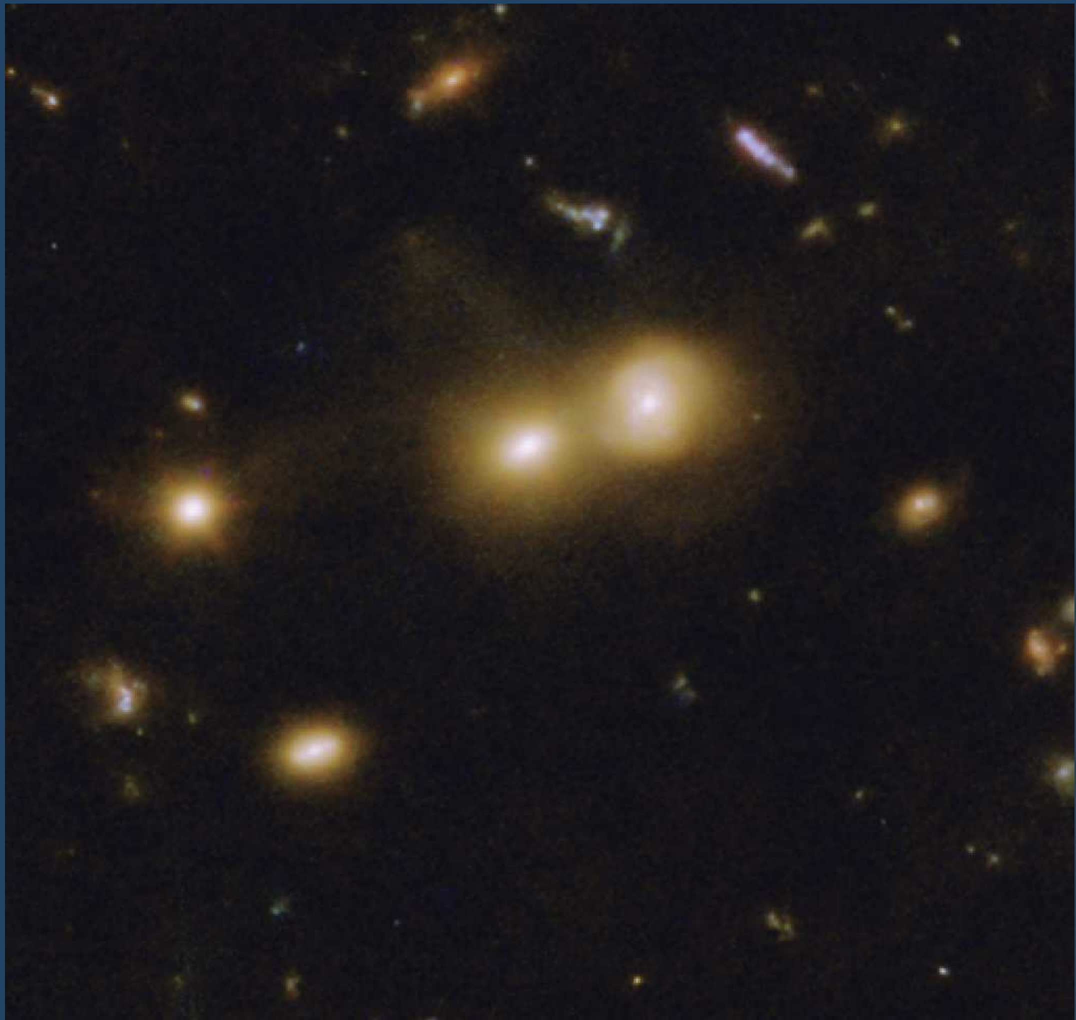

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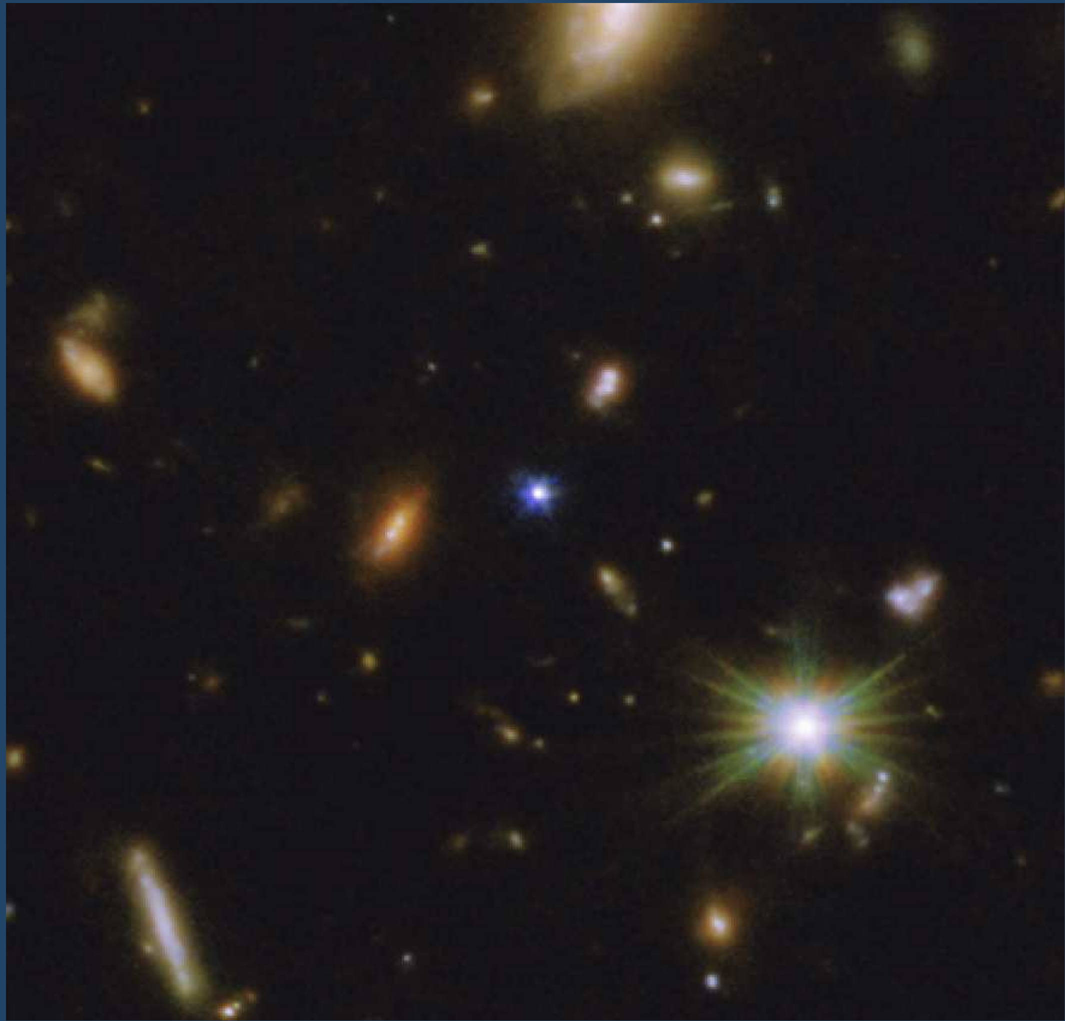

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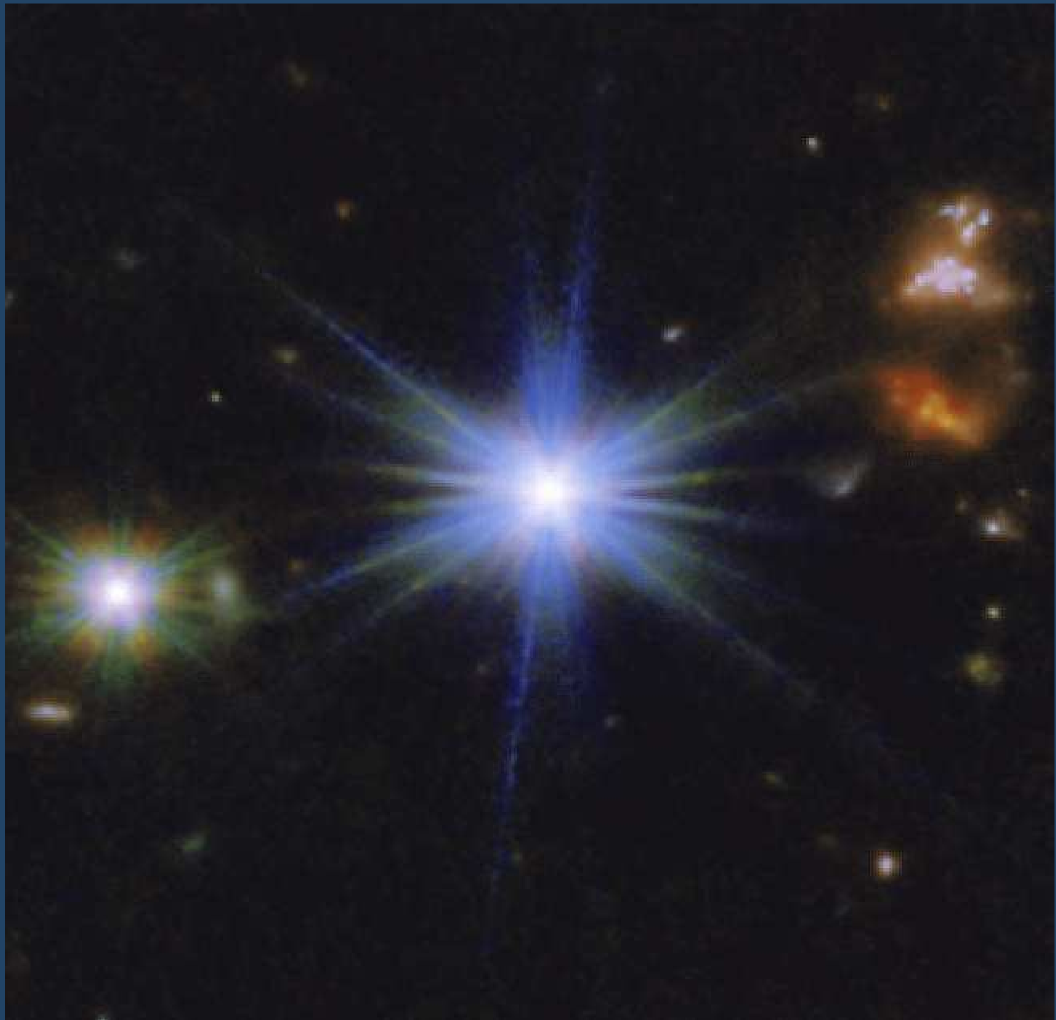 

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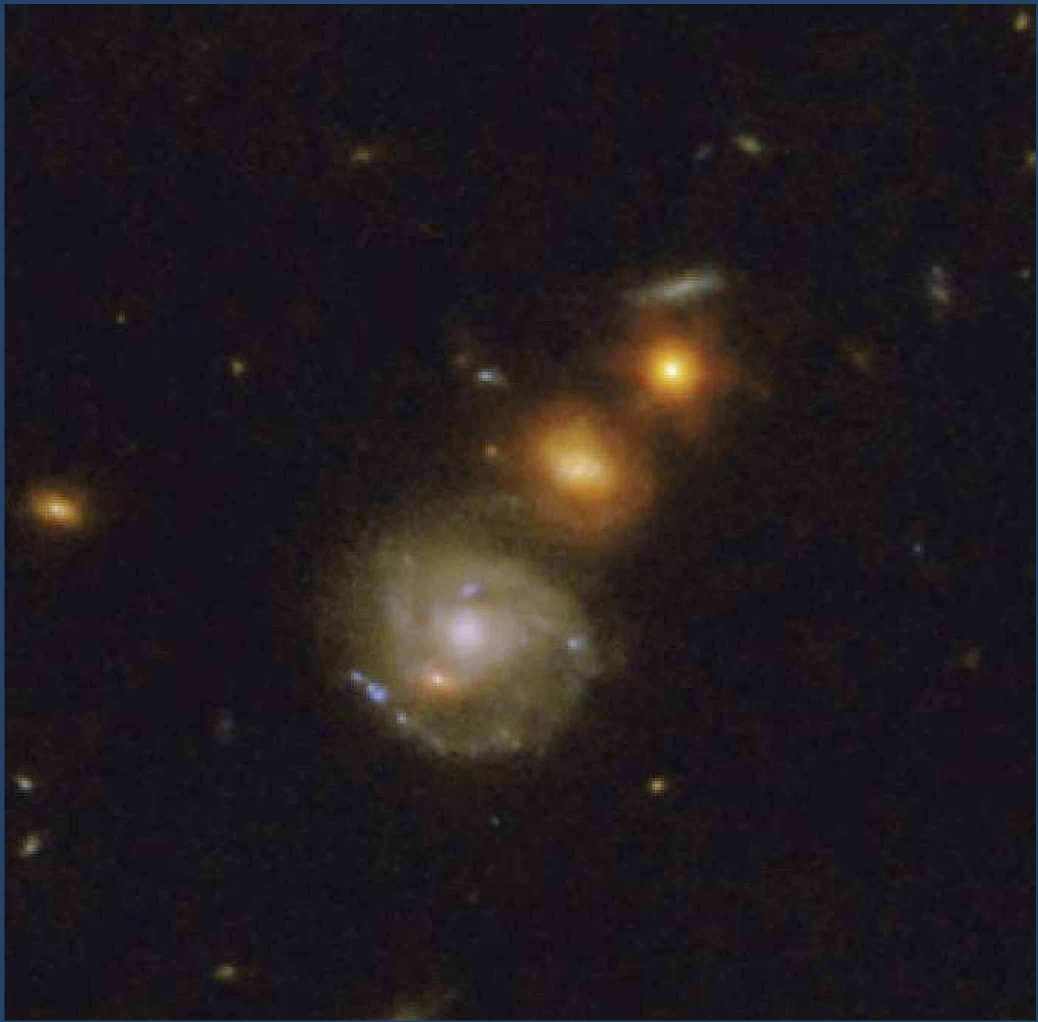

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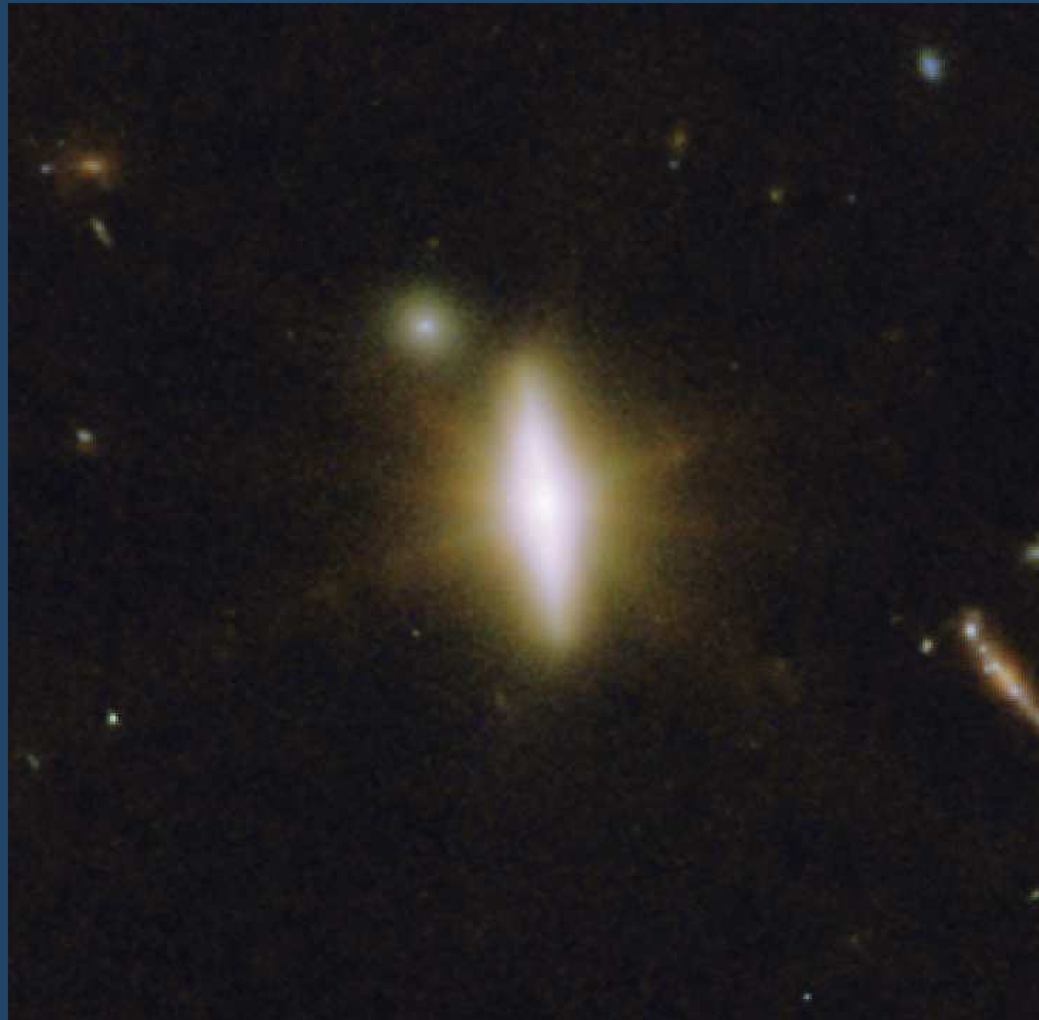

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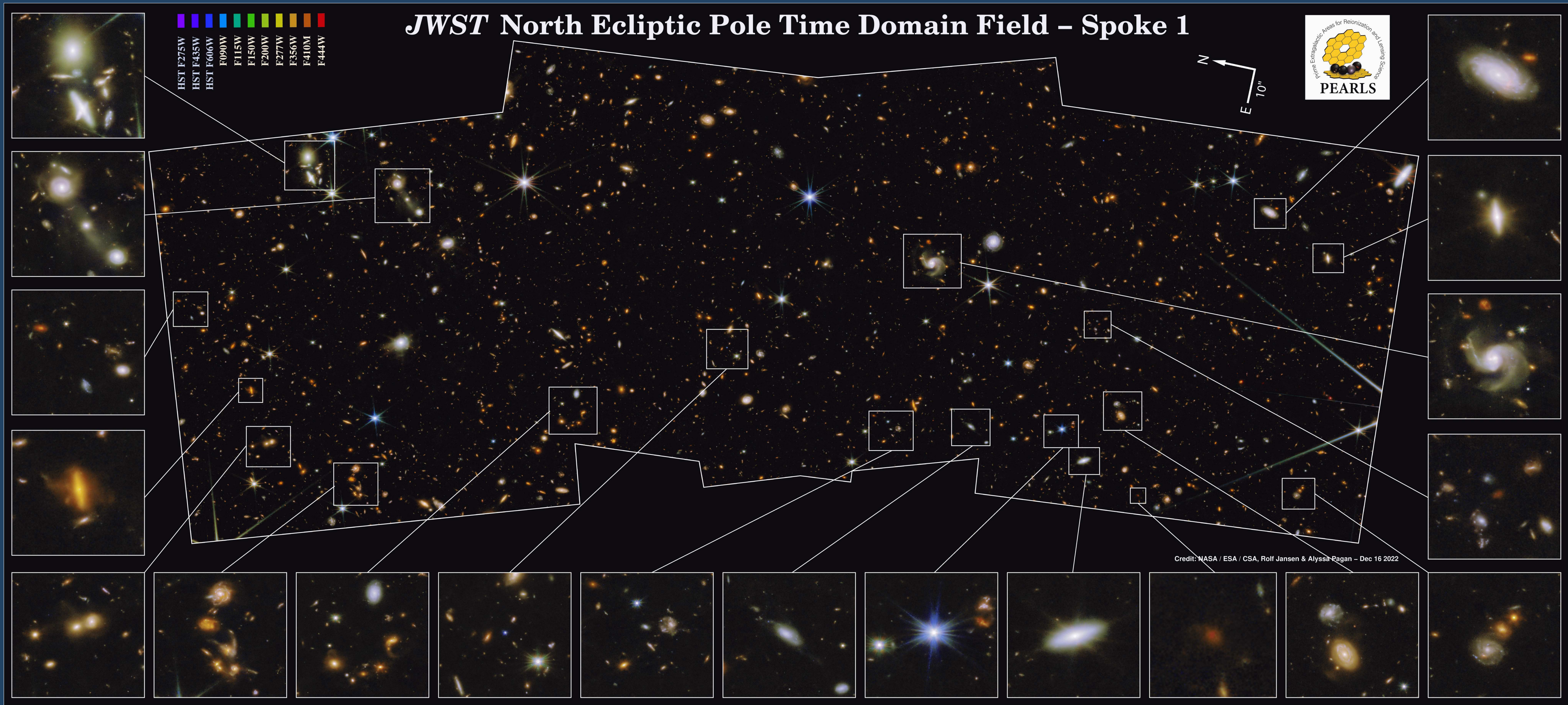

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